Columbia University
Department of Electrical Engineering
EE E4321. Problem Set #6. Adder design.
Due: November 6, 2025, 5 PM EST by electronic submission (see below)

Please carefully follow these instructions. Please submit your solutions to Problems 1 and 2 below as a single PDF file attachment in Courseworks. This will be submitted individually.
For Design Project Part I, please submit your writeup as a single PDF file attachment. This part is completed with your project partner and only one of your should submit this writeup, but please note clearly in the writeup the names of the two people who worked together on it.
Please attach the GDS layout file to your submission as well. To submit the layout, please stream out your design according to the instructions found on the class website.

First we'll do a few simple review problems.

1. Write a 7-bit two's complement representation of the following integers: 45, -38, 61, -57.

2. Perform the operations below in 6-bit two's complement notation:

001110 + 110010

010101 - 000111

000011 + 010101

111001 + 001010

010010 - 100101

Indicate if any overflows occur.

The next problem is part of your final design project; you should work on this with your design-project partner.

Design Project Part I. You are to design an 8-bit two's complement adder. Your adder has three inputs, $a < 0 : 7 >$, $b < 0 : 7 >$, and $subtract$, and two outputs, $sum < 0 : 7 >$ and $overflow$. When $subtract$ is 0, $sum = a+b$. When $subtract$ is 1, $sum = a - b$. You can design your adder using any architecture and circuit style you wish. Some of the possibilities we considered in class are:

- static ripple-carry adder
- static carry-bypass adder
- static carry-select adder
- static carry-lookahead adder

You could also do a domino carry-lookahead implementation. If you choose to do a domino implementation, you will have to be careful about monotonic signalling requirements and will want to do a separated latch design (to hide precharge times) when you put the whole microprocessor core together. To complete your adder design, I expect:

- Sized schematics
- Bit-stack layout of your adder design (remember to begin by making a tiling diagram and then stick layouts of your cells). We aware that the bit-stack you define here will be the bit-stack that you will have to use for the remainder of the components in your datapath.
- Spectre or Spectre FX results for the delay of the critical path
- Spectre FX runs to verify functionality